

Project Proposal: Meeting-to-Action Summarizer (Audio → Key Points, Tasks, Next Steps)

Team Members (5 students)

- Allantris Tellis – amt821
- Ashmit Mishra – am5087
- Mohnish Sao – ms4437
- Greyson Thibodeaux – gct74
- Addison Kyker – ask296

Scope of Project & Relation to AI

We will build a system that converts meeting or lecture audio into a concise brief containing: (1) an executive summary, (2) key points/decisions, and (3) action items with owners and due dates. The system combines speech recognition and LLM-based information extraction and formatting. Audio is first transcribed locally, then Gemini (via the Gemini API) is used to clean, structure, and validate outputs against a strict JSON schema to reduce hallucinations. We compare a deterministic baseline (regex/heuristics only) with the Gemini-enhanced pipeline to quantify the value of LLMs for meeting intelligence—clearly within the Generative AI/LLM scope.

Software / Models / Datasets (planned)

- ASR: faster-whisper (local Whisper) for transcription (CPU/GPU).
- LLM: Gemini API (e.g., Gemini 1.5) for summarization, key-point extraction, decisions, and action items (JSON output).
- Python libraries: numpy, pandas, dateparser, regex, tqdm; optional: pyannote (speaker diarization), matplotlib (plots).
- Frontend: Either Streamlit (Python) or React (JS).
- Data: short, consented recordings from team meetings/lectures; optionally small public samples for testing.
- Reproducibility: seedable runs; cached transcripts; fixed prompt templates; versioned JSON outputs.

Timeline

- Week 1: Data collection (short clips), ASR baseline working end-to-end; JSON output schema defined.

- Week 2: Integrate Gemini API; prompt templates for summary/decisions/actions; guardrails (low-confidence handling).
- Week 3: Build evaluation harness: action-item precision/recall/F1 vs. labeled gold; runtime/latency measurements.
- Week 4: Error analysis & improvements (owner/due-date parsing, ambiguous deadline handling; optional diarization).
- Week 5: Final polish: figures/tables, ablation (baseline vs Gemini), slides and short demo clip.

Division of Tasks

- Allantris Tellis (amt821): ASR pipeline & chunking; runtime profiling; caching.
- Ashmit Mishra (am5087): Gemini prompts & schema validation; ablations; error analysis.
- Mohnish Sao (ms4437): Data curation & annotation; report figures; slide deck & demo.
- Greyson Thibodeaux (gct74): Evaluation harness & metrics (P/R/F1); unit tests; batch processing.
- Addison Kyker (ask296): Frontend (Streamlit or React) & calendar export; packaging/CLI polish.